Diaspora Identity Bridge: Cross-Jurisdictional Identity Verification Without Central Authority

Decentralized Credentials for the Pars Network

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Abstract

We present ParsBridge, a decentralized identity verification system that enables members of the Persian diaspora to establish and prove identity claims across jurisdictional boundaries without relying on embassies, consulates, or centralized identity providers. ParsBridge leverages verifiable credentials (W3C VC), decentralized identifiers (DIDs), and zero-knowledge proofs to enable selective disclosure of identity attributes while preserving privacy. Our system introduces three novel credential types: *community attestations* (vouched by existing community members), *cultural knowledge proofs* (demonstrating cultural competency without revealing personal details), and *cross-jurisdictional credential chains* (linking identity documents from multiple countries without centralized verification). We formalize the trust model, prove that credential verification is sound and complete under the discrete logarithm assumption, and present results from a pilot deployment with 1,200 participants across 8 countries. ParsBridge achieves a verification success rate of 98.4% for legitimate credentials while detecting 100% of forged credentials in our adversarial evaluation.

1 Introduction

Identity verification is a foundational challenge for diaspora communities. Members of the Persian diaspora frequently need to prove aspects of their identity—nationality, education, professional qualifications, family relationships—across jurisdictional boundaries. Traditional identity verification relies on government-issued documents verified through diplomatic channels, a process that is inaccessible to many diaspora members due to:

- **Diplomatic barriers:** Iran's diplomatic relations with many host countries are limited or nonexistent, making consular verification difficult [Koser, 2007].
- **Safety concerns:** Engaging with Iranian consulates may expose diaspora members to surveillance or coercion, particularly political dissidents and activists [Dehghan, 2018].
- **Document inaccessibility:** Many refugees and asylum seekers lack access to their original identity documents, which may have been confiscated, lost, or never issued [UNHCR, 2019].
- **Credential recognition:** Educational and professional credentials from Iran are not universally recognized, requiring costly and slow evaluation processes [Nuffic, 2020].

1.1 Contributions

(i) **Credential Framework:** A W3C-compatible verifiable credential framework with three novel credential types designed for diaspora use cases.

(ii) **Community Attestation Protocol:** A decentralized attestation protocol where community members collectively vouch for identity claims, with Sybil resistance and collusion detection.

(iii) **Cultural Knowledge Proofs:** Zero-knowledge proofs of cultural competency that serve as soft identity signals without revealing personally identifiable information.

(iv) **Cross-Jurisdictional Chains:** A credential chaining mechanism that links identity documents from multiple countries, building cumulative trust without any single authoritative source.

(v) **Formal Security Analysis:** Proofs of soundness, completeness, and zero-knowledge properties of the credential verification protocols.

1.2 Threat Model

- **State-level adversary:** A government entity may attempt to identify, track, or de-anonymize diaspora members through the identity system.
- **Credential forgery:** Adversaries may attempt to create fraudulent credentials to access community resources or undermine trust.
- **Sybil attacks:** Adversaries may create multiple fake identities to manipulate the community attestation process.
- **Coercion:** Users may be compelled to reveal their credentials or identity information under duress.

2 Background

2.1 Decentralized Identifiers

Decentralized Identifiers (DIDs) [W3C, 2022a] are globally unique identifiers that do not require a centralized registration authority. A DID resolves to a DID Document containing public keys and service endpoints:

Definition 1 (Pars DID Method)**.** *A Pars DID has the form* `did:pars:<method-specific-id>` *where the method-specific-id is derived from the user's Ed25519 public key:*

$$\texttt{did:pars:}\|\mathsf{Base58}(\mathsf{BLAKE3}(pk)[:20]) \tag{1}$$

2.2 Verifiable Credentials

The W3C Verifiable Credentials Data Model [W3C, 2022b] defines a standard for expressing credentials in a way that is cryptographically secure, privacy-respecting, and machine-verifiable. A verifiable credential consists of:

- **Credential metadata:** Issuer DID, issuance date, expiration, schema.
- **Claim set:** Key-value pairs representing the credential's assertions.
- **Proof:** Cryptographic proof of integrity and issuer authenticity.

2.3 Zero-Knowledge Proofs

We use Schnorr-based zero-knowledge proofs [Schnorr, 1991] and BBS+ signatures [Boneh et al., 2004] for selective disclosure:

Definition 2 (BBS+ Signature)**.** *Given a public key* $pk = (g_1, g_2, h_0, h_1, \ldots, h_L) \in \mathbb{G}_1^{L+3}$ *and messages* $m_1, \ldots, m_L \in \mathbb{Z}_p$*, a BBS+ signature is a tuple* $(A, e, s) \in \mathbb{G}_1 \times \mathbb{Z}_p \times \mathbb{Z}_p$ *such that:*

$$A = \left(g_1 \cdot h_0^s \cdot \prod_{i=1}^{L} h_i^{m_i} \right)^{1/(e+sk)} \tag{2}$$

BBS+ signatures support efficient zero-knowledge proofs of signature possession with selective attribute disclosure, enabling a holder to prove possession of a signed credential while revealing only chosen attributes.

3 Credential Types

3.1 Type 1: Community Attestation

Community attestations allow existing community members to collectively vouch for a new member's identity claims.

Definition 3 (Community Attestation)**.** *A community attestation for claim c about subject s is a set of signed statements $\{(v_i, c, s, \sigma_i)\}_{i=1}^{k}$ where:*

- *v_i is the DID of voucher i (an existing community member with reputation $\geq \rho_{\min}$).*
- *c is the claim being attested (e.g., "subject holds a degree from University of Tehran").*
- *s is the subject's DID.*
- *σ_i is voucher i's BBS+ signature over $(c, s, timestamp)$.*
- *$k \geq k_{\min}$ (minimum attestation threshold, default 5).*

3.1.1 Attestation Protocol

Algorithm 1 Community Attestation Protocol

Require: Subject s with claim c, potential vouchers $V = \{v_1, \ldots, v_m\}$
Ensure: Valid community attestation or rejection

```
s publishes attestation request (c, s.did, evidence) to community
vouchers ← ∅
for v_i ∈ V willing to vouch do
    v_i reviews evidence and conducts verification (in-person or video)
    v_i assesses claim plausibility based on personal knowledge
    if v_i is satisfied then
        σ_i ← BBS+.Sign(v_i.sk, (c, s.did, timestamp))
        vouchers ← vouchers ∪ {(v_i, σ_i)}
    end if
end for
if |vouchers| ≥ k_min then
    weight ← Σ_{(v_i,σ_i)∈vouchers} reputation(v_i)
    if weight ≥ W_min then                                  ▷ Weighted threshold
        Create verifiable credential from attestations
        return APPROVED
    end if
end if
return INSUFFICIENT
```

3.1.2 Sybil Resistance

To prevent Sybil attacks on the attestation process, we employ several mechanisms:

1. **Reputation requirement:** Vouchers must have a minimum reputation score of $\rho_{\min}$, earned through sustained community participation.
2. **Social graph distance:** Vouchers must be at least 2 hops apart in the social trust graph to prevent collusion rings.
3. **Temporal diversity:** Vouchers' accounts must have been active for at least 6 months before they can attest.

4. **Stake requirement:** Vouchers stake a small amount of ASHA tokens that is slashed if the attested credential is later proven fraudulent.

3.2 Type 2: Cultural Knowledge Proof

Cultural knowledge proofs (CKPs) provide soft identity signals by demonstrating familiarity with Persian cultural knowledge that is difficult for outsiders to fake.

Definition 4 (Cultural Knowledge Proof)**.** *A CKP is a zero-knowledge proof of knowledge of correct answers to a challenge set drawn from a curated cultural knowledge base:*

$$\mathsf{CKP.Prove}(\mathit{answers}, \mathit{challenges}) \rightarrow \pi \tag{3}$$

such that a verifier can confirm:

$$\mathsf{CKP.Verify}(\pi, \mathit{challenges}) \rightarrow \{0, 1\} \tag{4}$$

without learning the specific answers.

3.2.1 Challenge Categories

Table 1: Cultural Knowledge Challenge Categories

Category	Questions	Difficulty	Example Topic
Language	200	Easy–Hard	Idiomatic expressions, proverbs
Literature	150	Medium–Hard	Poetry (Hafez, Rumi, Ferdowsi)
History	100	Medium	Historical events, figures
Geography	80	Easy–Medium	Cities, provinces, landmarks
Cuisine	60	Easy	Dishes, ingredients, customs
Music & Arts	70	Medium	Traditional instruments, artists
Nowruz	40	Easy	Traditions, Haft-sin items
Social customs	50	Easy–Medium	Tarof, hospitality norms
Total	**750**	–	–

3.2.2 ZK Challenge Protocol

Algorithm 2 Cultural Knowledge Proof Protocol

Require: Subject s, challenge set size $n = 20$, pass threshold $\theta = 0.75$
Ensure: CKP credential or rejection

```
challenges ← SampleChallenges(n)                ▷ Random, stratified by category
answers ← s.Answer(challenges)
hash_answers ← [BLAKE3(a_i || salt_i)]_{i=1}^n
s commits: C ← Commit(hash_answers)
Reveal salt values; verifier checks correct answer hashes
score ← |{i : hash_answers[i] = BLAKE3(correct_i || salt_i)}| / n
if score ≥ θ then
    Generate ZK proof: π ← ZKProof(score ≥ θ, C)
    Issue CKP credential with π (score range, not exact score)
end if
```

The ZK proof reveals only that the score exceeds the threshold, not the exact score or which questions were answered correctly. This prevents answer harvesting by adversaries.

3.3 Type 3: Cross-Jurisdictional Credential Chain

Credential chains link identity documents from multiple jurisdictions to build cumulative trust.

Definition 5 (Credential Chain)**.** *A credential chain $\mathcal{CH}$ for subject s is an ordered sequence of verifiable credentials:*

$$\mathcal{CH} = (VC_1, VC_2, \ldots, VC_k) \tag{5}$$

where each VC_i is issued by a different jurisdiction or institution, and adjacent credentials share at least one common attribute (name, date of birth, photo hash) that links them.

3.3.1 Chain Trust Score

The trust score of a credential chain is computed as:

$$\mathsf{Trust}(\mathcal{CH}) = \prod_{i=1}^{k} \mathsf{IssuerTrust}(VC_i) \cdot \prod_{i=1}^{k-1} \mathsf{LinkStrength}(VC_i, VC_{i+1}) \tag{6}$$

where $\mathsf{IssuerTrust}$ reflects the reliability of the issuing institution and $\mathsf{LinkStrength}$ measures the strength of the attribute linkage between adjacent credentials.

Table 2: Issuer Trust Levels

Issuer Type	**Trust Score**	**Example**
Government (recognized)	0.95	Canadian passport
Government (limited recognition)	0.70	Iranian national ID
Educational institution	0.85	University degree
Professional body	0.80	Medical license
Employer	0.75	Employment verification
Community attestation (5+ vouchers)	0.60	Identity claim
Cultural knowledge proof	0.40	Cultural competency

3.3.2 Chain Construction

Algorithm 3 Credential Chain Construction

Require: Subject s with credentials $\{VC_1, \ldots, VC_m\}$, target claim c
Ensure: Credential chain $\mathcal{CH}$ supporting claim c with maximum trust

```
G ← BuildCredentialGraph({VC_1, ..., VC_m})   ▷ Nodes = credentials, edges = shared attributes
Weight edges by LinkStrength
CH ← MaxTrustPath(G, VC_source, VC_target)   ▷ Modified Dijkstra with multiplicative weights
Verify all credentials in chain are valid and unexpired
Compute Trust(CH)
return CH
```

4 Selective Disclosure

4.1 BBS+ Based Selective Disclosure

ParsBridge uses BBS+ signatures for selective attribute disclosure:

Algorithm 4 Selective Disclosure Presentation

Require: Credential VC with attributes $\{a_1, \ldots, a_L\}$, disclosure set $D \subseteq [L]$
Ensure: Verifiable presentation revealing only $\{a_i : i \in D\}$

```
Parse BBS+ signature (A, e, s) from VC
Ā ← A · g_1^{r_1} for random r_1                    ▷ Randomize signature
Compute commitments for hidden attributes: C_j ← h_j^{a_j} · g_1^{r_j} for j ∉ D
Generate ZK proof π:
    Prove knowledge of (A, e, s, {a_j}_{j∉D}) such that:
        e(Ā, pk · g_2^e) = e(g_1 · h_0^s · ∏_{i∈D} h_i^{a_i} · ∏_{j∉D} C_j, g_2)
presentation ← (Ā, {a_i}_{i∈D}, {C_j}_{j∉D}, π)
return presentation
```

4.2 Predicate Proofs

Beyond selective disclosure, ParsBridge supports predicate proofs over hidden attributes:

- **Range proofs:** Prove age ≥ 18 without revealing exact date of birth.
- **Set membership:** Prove nationality $\in$ {Iran, Afghanistan, Tajikistan} without revealing which.
- **Equality:** Prove two credentials belong to the same person without revealing the linking attribute.

5 Security Analysis

5.1 Soundness

Theorem 1 (Credential Soundness)**.** *Under the q-SDH assumption in bilinear groups, no PPT adversary can produce a valid credential verification proof for a credential they do not possess, except with negligible probability.*

Proof. Soundness of the credential verification reduces to the unforgeability of BBS+ signatures under the q-SDH assumption [Boneh et al., 2004]. A successful forgery would imply the ability to produce a valid (A, e, s) tuple without knowledge of the signing key, which contradicts the q-SDH assumption. The selective disclosure proof inherits soundness from the underlying Schnorr-based ZK proof, which is sound under the discrete logarithm assumption. □

5.2 Zero-Knowledge

Theorem 2 (Zero-Knowledge Property)**.** *The selective disclosure protocol reveals no information about hidden attributes beyond what is logically implied by the disclosed attributes and the validity of the credential.*

Proof. We construct a simulator $\mathcal{S}$ that, given only the disclosed attributes $\{a_i\}_{i \in D}$ and the issuer's public key, produces presentations computationally indistinguishable from real presentations. The simulator chooses random $\bar{A} \in \mathbb{G}_1$, random commitments C_j, and simulates the ZK proof using the standard Schnorr simulator. Indistinguishability follows from the decisional Diffie-Hellman assumption in $\mathbb{G}_1$. □

5.3 Coercion Resistance

Property 1 (Deniable Credentials)**.** *ParsBridge supports deniable credentials: a user under coercion can present a "decoy" credential set that appears valid to the coercer but does not correspond to the user's actual identity, provided the coercer cannot independently verify the claims.*

This is implemented through a secondary key pair that the user can reveal under duress, which decrypts to a plausible but fake credential set stored alongside the real credentials.

6 Reputation System

6.1 Voucher Reputation

Voucher reputation is computed based on their attestation history:

$$\rho(v) = \rho_0 + \alpha \sum_i w_i \cdot \mathbf{1}[\text{attestation } i \text{ not challenged}] - \beta \sum_j w_j \cdot \mathbf{1}[\text{attestation } j \text{ challenged and upheld}] \tag{7}$$

where ρ_0 is the base reputation (earned through community participation), α is the reward for accurate attestations, β is the penalty for challenged attestations ($\beta > \alpha$ to discourage reckless vouching), and w_i is the weight of each attestation.

6.2 Reputation Decay

Reputation decays over time to ensure continued community participation:

$$\rho(v, t) = \rho(v, t_0) \cdot e^{-\lambda(t - t_0)} + \Delta\rho(t_0, t) \tag{8}$$

where λ is the decay rate (calibrated so reputation halves in 1 year without activity) and $\Delta\rho(t_0, t)$ is the reputation earned during $[t_0, t]$.

7 Implementation

7.1 Technology Stack

Table 3: Implementation Technology

Component	Technology	Rationale
DID method	Custom (did:pars)	Pars-specific requirements
Credential format	W3C VC JSON-LD	Interoperability
Signatures	BBS+ (BLS12-381)	Selective disclosure
ZK proofs	Groth16 (Circom)	Compact proofs
Storage	IPFS + local	Decentralized persistence
Resolver	Pars DHT	DID resolution
Client	React Native + Rust FFI	Cross-platform

7.2 Wallet Application

The ParsBridge wallet is a mobile application providing:

- Credential storage in hardware-backed secure enclave.

- Biometric-protected access (fingerprint/face).
- QR code-based credential presentation.
- NFC-based proximity verification for in-person scenarios.
- Offline verification capability (cached issuer public keys).

8 Evaluation

8.1 Pilot Deployment

We conducted a pilot deployment with 1,200 participants across 8 countries:

Table 4: Pilot Deployment Demographics

Country	Participants	Credentials	Attestations	CKPs
United States	320	1,240	480	290
Canada	240	890	360	210
Germany	180	640	270	160
United Kingdom	120	430	180	100
Sweden	100	350	140	85
Australia	80	280	110	70
Turkey	100	310	120	80
UAE	60	190	70	45
Total	**1,200**	**4,330**	**1,730**	**1,040**

8.2 Verification Accuracy

Table 5: Verification Results

Credential Type	Legitimate (%)	Forged Detected (%)	Avg. Time (s)
Community attestation	97.8	100.0	0.8
Cultural knowledge proof	94.2	100.0	12.4
Credential chain	98.4	100.0	2.1
Selective disclosure	99.9	100.0	0.3
Overall	**98.4**	**100.0**	–

8.3 Cultural Knowledge Proof Analysis

Table 6: CKP Performance by Population

Population	Mean Score	Pass Rate ($\geq$75%)	Std Dev
Born in Iran, <10 yr diaspora	91.2%	98.4%	5.3%
Born in Iran, 10–20 yr diaspora	84.7%	94.1%	8.1%
Born in Iran, >20 yr diaspora	78.3%	82.6%	10.4%
Born in diaspora, Persian household	72.1%	71.3%	12.8%
Non-Persian control group	31.4%	2.1%	14.2%

8.4 Sybil Attack Resistance

We simulated Sybil attacks with varying strategies:

Table 7: Sybil Attack Detection Results

Attack Strategy	Fake IDs	Detected	Detection Rate	False Pos.
Random identities	100	100	100%	0%
Coordinated ring (5 nodes)	50	48	96%	0.2%
Compromised vouchers (2)	20	19	95%	0.5%
Slow infiltration (6 mo.)	10	8	80%	0.3%

8.5 Comparison with Existing Systems

Table 8: Comparison with Identity Systems

Feature	ParsBridge	euPort	Sovrin	WorldID	UNHCR ID
Decentralized	Yes	Yes	Partial	No	No
Selective disclosure	BBS+	No	CL sigs	No	No
No central authority	Yes	Yes	Partial	No	No
Cultural proofs	Yes	No	No	No	No
Credential chains	Yes	No	No	No	No
Coercion resistant	Yes	No	No	No	No
Offline verify	Yes	No	No	No	Yes
Diaspora-specific	Yes	No	No	No	Partial

9 Use Cases

9.1 Employment Verification

A Persian engineer in Germany needs to prove their University of Tehran degree to an employer without contacting Iranian authorities:

1. Subject presents a credential chain: Iranian university transcript (CKP-backed) → credential evaluation by WES Canada → German qualification recognition.
2. Selective disclosure reveals only the degree title and field, not grades, student ID, or personal details.
3. Community attestation from 5 alumni of the same university provides additional backing.

9.2 Financial Services

A diaspora member needs KYC verification for a Pars Network financial service:

1. Subject presents: host country residence proof (government-issued) + credential chain to Iranian identity.
2. Predicate proof: age ≥ 18 (without revealing birthdate).
3. Set membership proof: nationality $\in$ {Iran, Afghanistan, Tajikistan} (without revealing which).

9.3 Family Reunification

A refugee needs to prove family relationship for immigration purposes:

1. Subject presents: community attestations from 5+ community members who know both family members.
2. Cross-reference: shared CKP results (family-specific cultural knowledge).
3. Credential chain: any available official documents from both origin and host countries.

10 Security Analysis

10.1 Sybil Resistance

ParsBridge resists Sybil attacks through multiple mechanisms:

Theorem 3 (Sybil Cost)**.** *Creating k Sybil identities in ParsBridge requires attestations from $\geq 3k$ distinct verified community members and $\geq k$ independent CKP sessions, assuming no attestor vouches for more than one Sybil.*

Proof. Each identity requires ≥ 3 independent community attestations (Section 5). Attestors stake ASHA tokens and are penalized for vouching for fraudulent identities. The CKP component requires interactive cultural knowledge verification that cannot be delegated without revealing answers, creating a distinct bottleneck per Sybil. Therefore, k Sybils require at minimum $3k$ unique attestors and k CKP sessions, each requiring domain knowledge and interactive participation. □

10.2 Collusion Resistance

If a group of m malicious attestors collude to vouch for fake identities:

1. **Reputation stake:** Each attestor stakes 50 ASHA per attestation. Slashing for confirmed fraud removes 5× the stake.
2. **Social graph analysis:** Attestation patterns are analyzed for anomalies (e.g., the same group repeatedly vouching for each other). Suspicious clusters trigger community review.
3. **CKP diversity:** CKP questions are drawn from a pool of 500+ questions across 12 cultural domains. Sharing answers for more than 2–3 identities would require memorizing hundreds of culturally specific responses.

10.3 Privacy Under Device Compromise

If an adversary seizes a device containing ParsBridge credentials:

- Credential private keys are encrypted at rest using a user-derived key (PIN + biometric on supported devices).
- BBS+ signatures support selective disclosure: the adversary cannot learn attributes beyond what the user explicitly reveals.
- Credential usage history is not stored on-device; the adversary cannot determine which verifications were performed.
- Emergency revocation: the user can revoke all credentials from any device using their social recovery network.

11 Limitations and Future Work

1. **Cold Start:** New community members without existing connections face bootstrapping challenges. Integration with established diaspora organizations could help.
2. **Cultural Bias:** CKPs may disadvantage diaspora members with less exposure to mainstream Persian culture. We are working on region-specific and generational variants.
3. **Legal Recognition:** ParsBridge credentials are not yet legally recognized. Partnerships with credential evaluation agencies are in progress.
4. **Scalability:** BBS+ proof generation is computationally expensive on low-end devices. Delegated proof generation with trusted proxies is being investigated.
5. **Revocation:** Credential revocation currently uses a revocation registry on the Pars DHT. More efficient accumulator-based revocation is planned.
6. **Interoperability:** Federation with non-Persian identity systems (Kurdish, Armenian, Afghan) for shared regional identity verification is under design.

12 Deployment Considerations

12.1 Credential Issuance Ceremonies

Initial credential issuance requires in-person verification at community gatherings. We designed a ceremony protocol that combines cultural events with identity bootstrapping:

Algorithm 5 Community Credential Ceremony

Require: Community gathering with ≥ 10 verified members, new participant P
Ensure: P receives initial ParsBridge credential

```
P presents any available identity documents (passport, national ID, birth certificate)
k ≥ 3 verified community members independently attest to P's identity
Each attester a_i provides:
    Cultural knowledge verification (conversational assessment)
    Community relationship attestation (how they know P)
    Time-stamped signed statement recorded on mobile device
Ceremony coordinator aggregates attestations
if k ≥ 3 attestations consistent and documents present then
    Issue Level 2 credential to P
else if k ≥ 5 attestations consistent (no documents) then
    Issue Level 1 credential to P
end if
P generates DID keypair on personal device
Credential bound to P's DID and recorded on Pars DHT
```

Over 18 months, 47 credential ceremonies were held across 12 cities (Los Angeles, London, Berlin, Toronto, Sydney, Stockholm, Vienna, Istanbul, Dubai, Kabul, Dushanbe, Paris), issuing 1,200 initial credentials. Average ceremony duration: 3 hours with 15–40 participants.

12.2 Integration with Existing Systems

ParsBridge credentials can be used to:

Table 9: Integration Points for ParsBridge Credentials

System	Credential Use	Status
Pars governance (ParsDAO)	Voter eligibility verification	Live
ParsArchive curation	Curator admission proof	Live
Translation bounties	Translator language attestation	Live
Community mutual aid	Membership verification	Pilot
Educational access	Student credential bridging	Planned
Employment	Professional credential verification	Planned

12.3 Credential Recovery

Lost credentials can be recovered through the social recovery mechanism described in **?**. A minimum of 3 out of 5 designated guardians can authorize credential re-issuance to a new DID, preserving the attestation history while binding to fresh cryptographic material.

13 Conclusion

ParsBridge provides a practical decentralized identity system for the Persian diaspora, enabling cross-jurisdictional identity verification without reliance on embassies, consulates, or centralized identity providers. By combining community attestations, cultural knowledge proofs, and credential chains with strong cryptographic privacy guarantees, ParsBridge empowers diaspora members to establish and prove their identity in a privacy-preserving, coercion-resistant manner. Field deployment across 8 countries with 1,200 participants demonstrates both technical feasibility and community acceptance of the decentralized identity model.

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